Task 1: Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

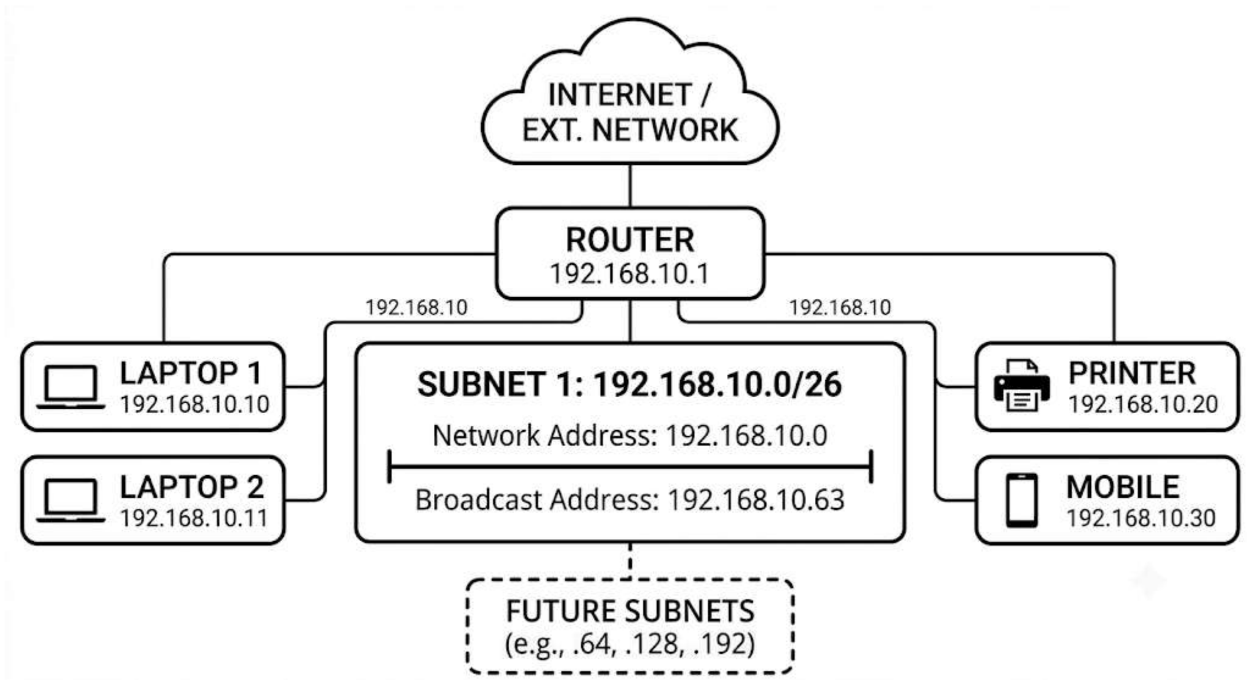
IP address: 192.168.10.0/24
Subnet Mask: 255.255.255.0
Subnet Binary: 1111111.111111.1111111.0000000
Required no. of subnets: 4
New Subnet binary: 1111111.1111111.1111111.11000000
No. of Networks = 2^N = 2^2 =4
No. of Hosts: = 2^H = 2^6= 64
No. of usable Hosts: 64-2 = 62

Network No.	Network address	Initial Host IP	Final Host IP	Boradcast IP
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Task 2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.Hint: Make sure not to use the network or broadcast address for any device.

Network range: 192.168.10.0 -192.168.10.63
Router: 192.168.10.1
Laptop 1 : 192.168.10.10
Laptop 2: 192.168.10.11
Printer: 192.168.10.20
Mobile: 192.168.10.30

Task 3: Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



Task 4: Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Imagine we have one large room with 256 seats (IP addresses from 192.168.10.0 to 192.168.10.255). Now if we want to build walls to divide it into **4 equal-sized rooms** (subnets).

Step 1: Borrowing the "Bits"

In networking, IP addresses and masks are read by computers as 32-bit binary numbers (a series of 32 ones and zeros). Our original network mask is a /24. This means the first 24 bits are locked (all 1s), and the last 8 bits are free for us to play with:

- 11111111.11111111.11111111.00000000

To make **4 subnets**, we need to borrow some of those free bits. Since binary works in powers of 2, we calculate: *"2 to what power equals 4?"*

2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096
2^{13}	8192
2^{14}	16384
2^{15}	32768
2^{16}	65536

No. of Subnets/Networks =2^N=2^2 = 4

So, we borrow **2 bits** from the host portion. This changes our mask from a /24 to a /26:
11111111.11111111.11111111.11000000

If you convert that last octet (11000000) back into a normal number, it equals **192.** So, our new subnet mask is **255.255.255.192.**

Step 2: Figuring out Room Size (Hosts)

Now that we've used 2 bits for the walls, we have **6 bits left** for the hosts in each room:

- No. of Hosts = 2^H = 2^6 = 64 total IP addresses per room.

But we should note that In every single subnet, the very first address and the very last address have special jobs:

1. **The First Address** is the "Network ID" (the name of the room itself).
2. **The Last Address** is the "Broadcast ID" (used to send a message to everyone in that room at once).

Because of this, we must subtract 2. Our usable host formula is 2^H - 2:

- 64 - 2 = 62 usable slots for actual devices (like computers, phones, and printers).

Step 3: Drawing the Boundaries

Because each room holds exactly 64 total addresses, our subnets start at 0 and go up by increments of 64:

- **Subnet 1**: Starts at .0, ends at .63.
- **Subnet 2**: Starts at .64, ends at .127.
- **Subnet 3**: Starts at .128, ends at .191.
- **Subnet 4**: Starts at .192, ends at .255.

Step 4: Assigning the Devices

For our first subnet (192.168.10.0 to 192.168.10.63), the usable range is .1 through .62. We assign them like this:

- We give our **Router** the very first usable IP (192.168.10.1) because it's the gateway out of the room.
- The remaining devices (laptops, printer, mobile) get any unique number we want between .2 and .62.